# MATHEMATICS REFLECTION WORK

## Reflection Work 2

## 1. Reflection Topic

**Exploring Trigonometric Ratios and their Real-Life Applications (Heights and Distances).**

## 2. Objectives of the Reflection

- To reflect upon the fundamental concepts of trigonometric ratios.
- To apply trigonometric principles (sine, cosine, tangent) to calculate unknown heights and distances in real-world scenarios.
- To understand and differentiate between the **Angle of Elevation** and the **Angle of Depression**.
- To appreciate how abstract mathematical concepts solve practical physical problems.

## 3. Pre-requisite Knowledge

Trigonometry is based on the relationships between the sides and angles of a right-angled triangle. The primary ratios we use are:

- $\sin \theta = \text{Opposite} / \text{Hypotenuse}$
- $\cos \theta = \text{Adjacent} / \text{Hypotenuse}$
- $\tan \theta = \text{Opposite} / \text{Adjacent}$

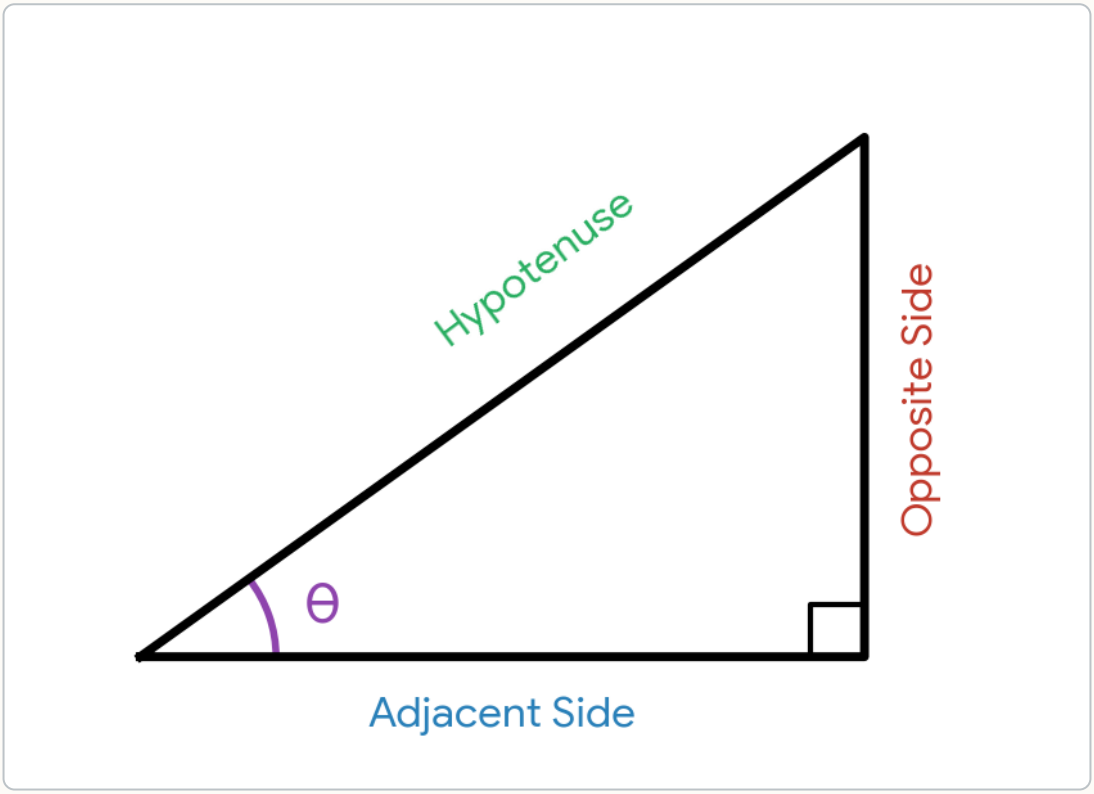



*Fig 1: Identifying the sides of a right-angled triangle relative to angle θ*

## 4. Reflections and Real-Life Applications

### Reflection 1: Finding Heights using the Angle of Elevation

When an observer looks at an object situated above their horizontal line of sight, the angle formed is the **Angle of Elevation**.

> ***Situation:*** *I imagined standing 10 meters away from a tall tree. Looking up at the top of the tree, the angle of elevation is exactly* ***45°****. How tall is the tree?*

Using the tangent ratio:

$$\tan 45° = \text{Height } (h) / \text{Distance } (d)$$

$$1 = h / 10 \quad \Rightarrow \quad h = 10 \text{ meters}.$$

*My Observation:* I realized that whenever the angle of elevation is 45°, the height of the object is exactly equal to the distance from its base! This is a fascinating and highly useful shortcut.

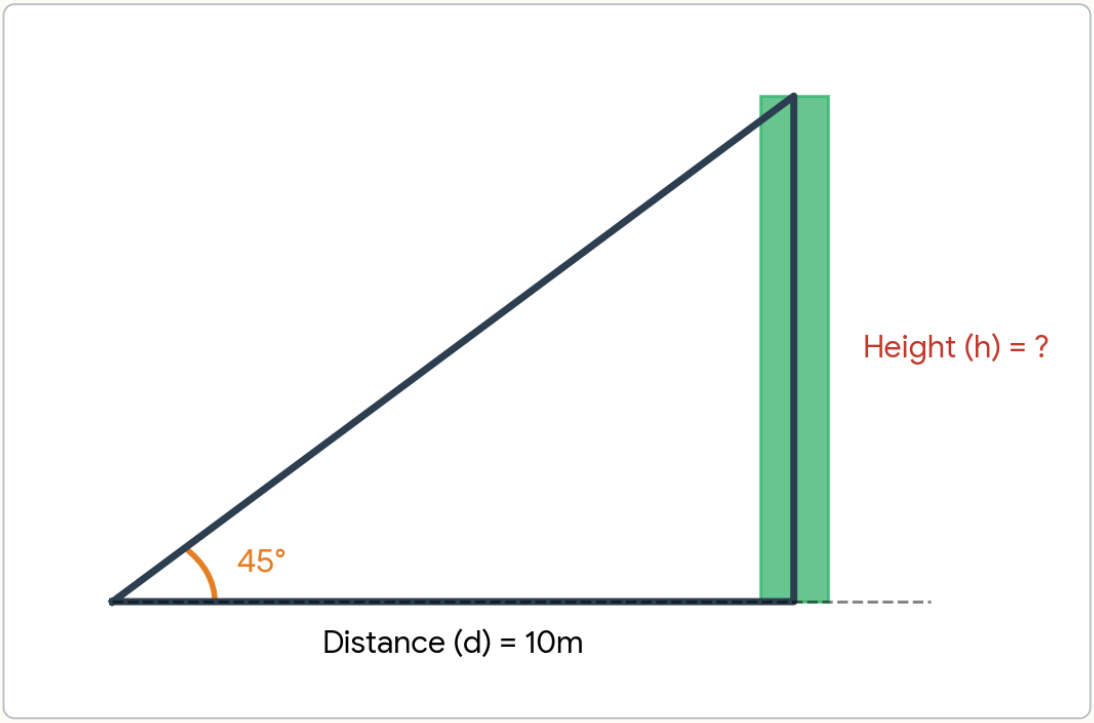



*Fig 2: Calculating height using the Angle of Elevation*

## Reflection 2: Finding Distances using the Angle of Depression

When an observer looks down at an object below their horizontal line of sight, the angle formed is the **Angle of Depression**. Geometrically, this angle is equal to the angle of elevation from the object to the observer due to alternate interior angles.

> ***Situation:*** *An observer on top of a 50m high lighthouse spots a boat. The angle of depression is* $30°$*. How far is the boat from the base of the lighthouse?*

Using the tangent ratio:

$\tan 30° = \text{Height} / \text{Distance (d)}$

$1 / \sqrt{3} = 50 / d \quad \Rightarrow \quad d = 50\sqrt{3}$ meters (approx. 86.6 meters).

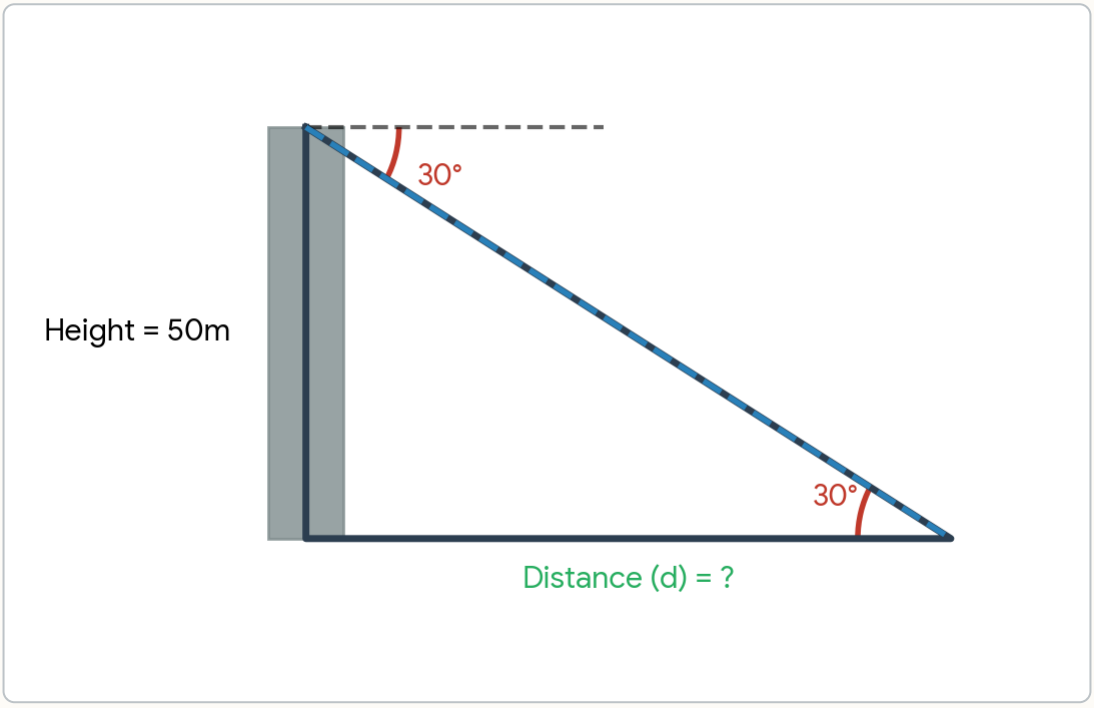



*Fig 3: Calculating distance using the Angle of Depression*

## 5. Key Observations & Confusion Clarification

- **Observation:** Trigonometric ratios magically connect angles (in degrees) to linear measurements (in meters or centimeters). Without physically climbing a mountain or a building, we can accurately find its height.
- **Common Confusion:** While solving problems, it is common to get confused about where to mark the angle of depression. I learned that it must always be drawn from the *horizontal line of sight* downwards, not from the vertical structure.

## 6. Conclusion

This reflection work helped me realize that Trigonometry is a highly practical branch of mathematics. The concepts of Heights and Distances are actively used in architecture, astronomy, navigation, and engineering. By understanding how to model a real-world scenario into a simple right-angled triangle, complex measurement problems become very easy to solve.

## 7. Reference Books Used

- 10th Class Mathematics Textbook (Telangana State Board)

## 8. Acknowledgements

I would like to express my sincere gratitude to my mathematics teacher for guiding me through the concepts of Trigonometry, which made this reflection work possible.